

Hayden Choi

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Education

McMaster University

Expected May 2028

Honours Computer Science Co-op, Bachelor of Applied Science (GPA: 3.9/4.0)

Hamilton, ON

- Relevant Coursework: Data Structures & Algorithms, Software Development, Distributed Databases, Operating Systems

Experience

Shopify

Jan 2027 – Apr 2027

Software Engineer Intern

Toronto, ON

- Incoming Winter 2027

Wealthsimple

Sept 2026 – Present

Software Engineer Intern

Toronto, ON

- Deployed a customer-facing risk event API in the **Rails** monolith, streaming device session signals through **Kafka** into the fraud decision platform for real-time evaluation, preventing **\$1.3M+** in quarterly account takeover losses
- Integrated mobile telemetry in **Kotlin** emitting call and screen-capture flags synchronously at transaction submit across **4M+** client sessions, enforcing a fail-closed consumer contract that escalates dropped detections instead of passing them
- Shipped composite rules scoring device context against transaction behaviour, focal-blocking **3** money movement flows
- Deprecated a legacy bank linking decision service, removing **14** call sites reducing decision platform latency by **12%**

Trend Micro

May 2026 – Aug 2026

Software Engineer Intern

Ottawa, ON

- Built a behavioral ranking framework scoring **1,500+** risk event types on real fleet remediate-vs-dismiss outcomes, mining **8M+** human action decisions from **2,000+** enterprise tenants on a **Trino/Iceberg** lakehouse
- Designed a cross-repo **CI/CD** pipeline cutting the **5 day** risk event definition onboarding down to **2 days**
- Shipped a credential-free **EKS** service across **12** AWS regions on **Pod Identity**, owning its **Terraform** footprint

Knockri

Dec 2025 – Apr 2026

Software Engineer Intern

Toronto, ON

- Created an AI agent to autonomously resolve support tickets, cutting inbox volume by **83%** and support costs by **61%**
- Implemented a **Heartbeat API** using **GraphQL/UUID4** to restrict **7,000+** duplicate launches via backend locking

Brighter Signals BV

May 2025 – Aug 2025

Software Engineer Intern

Markham, ON

- Developed **5+** specialized prototypes for investor demos which secured **\$1.6M+** in funding for **MVP** development
- Co-authored an **IROS 2025** paper and named a **Best Poster Finalist** for research on data-rich tactile sensors

Mana Immigration

May 2024 – Sept 2024

Software Engineer Intern

Markham, ON

- Architected **6+** full-stack Flutter apps for international clients using a Factory Pattern for reusable UI components

Projects

WatchTower | CSE Challenge Winner, Hack the North 2026

Devpost | GitHub

FastAPI, React, TimescaleDB, scikit-learn, LLMs

- Built a hybrid detector pairing **6** rules with an **Isolation Forest** to correlate **180K+** HTTP logs into incidents
- Cut event query latency **96%** (79.7ms to 2.9ms) by redesigning server-side pagination on **TimescaleDB** hypertables

Proteus | 2nd Overall & Best Use of DigitalOcean, cuHacking7

Devpost | GitHub

FastAPI, Next.js, MongoDB Atlas Vector Search, Redis, DigitalOcean

- Created a genomic IDE turning plain-English goals into scored DNA with predicted folds, accelerating wet lab testing
- Built the literature **RAG** layer, embedding **PubMed** abstracts for semantic search gated to model-edited regions

ColdApproach AI

GitHub

Next.js, FastAPI, AWS (SES, SQS, Lambda), OpenAI, Hunter.io

- Built an AI outreach platform with **FastAPI** and **Next.js** using **Hunter.io** discovery and a two-pass **GPT** pipeline
- Engineered an async batch pipeline with **AWS SQS** to process up to **20** URLs through scraping and generation stages

Technical Skills

Languages: Python, TypeScript, Ruby, Java, Kotlin, JavaScript, C++, SQL

Frameworks & Tools: Docker, Kubernetes, Ruby on Rails, FastAPI, React, GraphQL, Terraform, Linux, Git, CI/CD

Data & Cloud: AWS (S3, Glue, Athena), Apache Kafka, Trino, Apache Iceberg, PySpark